



UNIVERSITY EXAMINATIONS: 2013/2014
ORDINARY EXAMINATION FOR THE BACHELOR OF SCIENCE
IN INFORMATION TECHNOLOGY & BUSINESS INFORMATION
TECHNOLOGY

BBIT 103 BIT 1305 COMPUTER NETWORKS

DATE: AUGUST, 2014

TIME: 2 HOURS

INSTRUCTIONS: Answer Question ONE and any other TWO

QUESTION ONE: 30 MARKS-COMPULSORY

- (a)(i) Distinguish between CSMA/CD and CSMA/CA. [4 Marks]
(ii) Where are they applied? [2 Marks]
- (b) Briefly describe **two** similarities and **two** differences between switches and routers. [4 Marks]
- (c) Briefly discuss the **three (3)** fundamental characteristics of a data communication system. [3 Marks]
- (d) Briefly discuss any **three (3)** benefits of the OSI Reference Model. [6 Marks]
- (e)(i) Which class does the IPv4 address **130.248.45.89** belong to? Show your working. [2 Mark]
(ii) What is the Network ID for this IP address? [1 Mark]
- (f) Contrast the following:
i). Circuit switching and Packet switching. [2 Marks]
ii). Single mode and Multimode fiber. [2 Marks]
- (g) A network topology is a physical layout of the network. Discuss the following topologies, using appropriate diagrams: **Bus** and **Ring**. [4 Marks]

QUESTION TWO

- (a) List **four (4)** functions that are performed by a Network Interface Card (NIC).
[4 Marks]
- (b) As a network administrator, you have two options when it comes to assigning IP addresses to the hosts in your network. Briefly explain:
 - (i) When it is appropriate to use **Static IP configuration** option.
[4 Marks]
 - (ii) When it is appropriate to use **Dynamic IP configuration** option.
[4 Marks]
- (c) Discuss briefly **four (4)** factors that must be taken into account when deciding what type of Network Operating System to install on a network. [8 Marks]

QUESTION THREE

- (a) You and your friend wish to transfer huge files between your two laptops. It would take a lot of time to accomplish this task if you decided to use a memory stick. Your friend asks you to set a simple peer-to-peer network to achieve the objective.
 - (i) What equipment do you require to accomplish this task? [4 Marks]
 - (ii) What type of transmission media would you use? [1 Mark]
 - (iii) What type of cable will you construct? [1 Mark]
 - (iv) Which network troubleshooting commands are you likely to use in this task? [2 Marks]
 - (v) How will you set up the simple peer-to-peer network of the two laptops? [2 Marks]
 - (vi) What configurations are necessary for this task to be accomplishment? [4 Marks]
- (b) Differentiate between the following terminologies as used in computer networks:
 - (i) Broadband and Baseband signaling. [2 Marks]
 - (ii) Half duplex and full duplex transmission. [2 Marks]
 - (iii) Impulse and Thermal noise. [2 Marks]

QUESTION FOUR

- (a) Internet services often use the so-called client server paradigm. New ad-hoc networks, on the other hand, use peer-to-peer services. Compare these two ways

- and give examples of services or applications. Explain the benefits, drawbacks, and problems of both paradigms. [6 Marks]
- (b) List and briefly explain any **three (3)** things that computer networks provide to people. [6 Marks]
- (c) Describe the steps a client goes through to get an IP address from a DHCP server. [6 Marks]
- (d) Explain any **two (2)** possible sources of noise in cables. [2 Marks]

QUESTION FIVE

- (a) Explain how wireless LANs communicate. [6 Marks]
- (b) Explain why the Four Repeater Rule is important. [3 Marks]
- (c) Outline the **five (5)** components of a data communication system. [5 Marks]
- (d) Describe briefly the three components of a fiber optic cable. [3 Marks]
- (e) Briefly discuss any **two (2)** functions of a Network Administrator. [3 Marks]